Lord of the Wings [2026]

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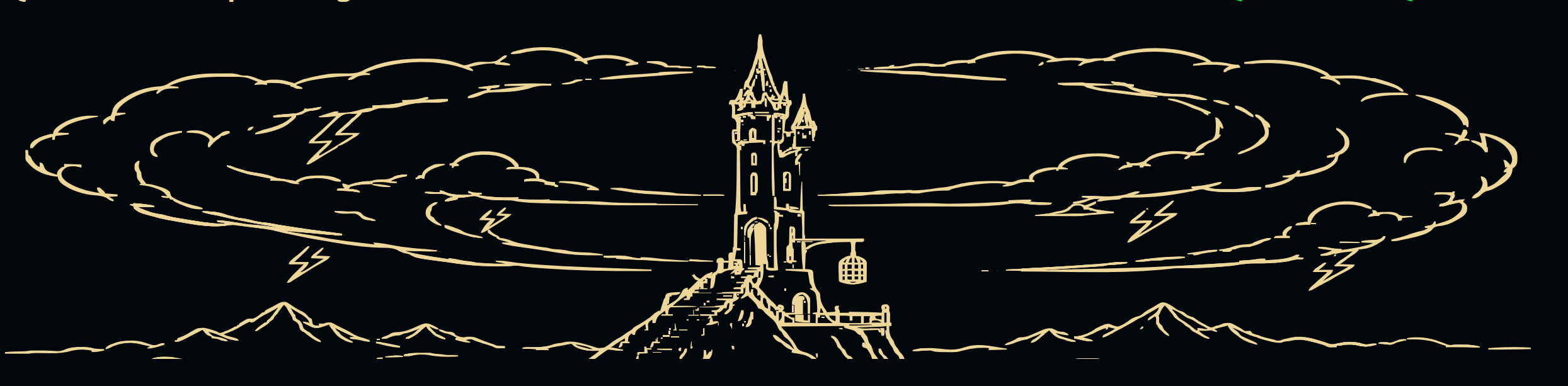

Part I

Story section

The analysis of the innkeeper's records revealed that the contamination had spread much further than we had feared. It was not only present in the bread, but also in meat pies, bowls of pottage and, most alarmingly, the kingdom's extremely popular turnip stew.

I decided that we needed the help of someone who understood dark magic far better than any of us. Fortunately, an old friend of mine from the Academy of Magic lived nearby. His name was Sarumonk. We had not spoken for several years, but I was certain that he would remember me.

He did. Unfortunately, by the time we reached his tower, the corruption had already found him. Sarumonk appeared at the highest window, surrounded by black smoke and crackling sparks, and immediately accused me of ruining his final year at the academy. He had apparently never forgiven me for enchanting his graduation robes to shout, "I copied this spell from the textbook!" whenever he raised his wand.

Before I could explain that it had seemed very funny at the time, Sarumonk trapped the rest of the team inside a large iron cage hanging above the tower courtyard. He then challenged me to a lightning duel. There was no time to argue. Sparks were already gathering around the tip of his staff.

I tried to remember the rules of a lightning duel. The aim was to summon lightning from the clouds above the battlefield. To do this, a wizard first had to excite one of the clouds by adding a suitable amount of magical charge to it.

The charge could then travel from one cloud to another. Each cloud could pass part of the charge to several other clouds, but **every connection worked in only one direction.** A cloud could pass charge to another cloud without being able to receive it back through the same connection.

I remembered a convenient way to record the clouds and their connections:

```
id=ID, energy=ENERGY, connections=ID_1,ID_2,...
```

The `id` was the cloud's unique identifier, while `energy` described its initial magical energy. The identifiers listed after `connections` were the clouds to which it could pass charge directly.

Before facing Sarumonk, I decided to practise the entire process on a small example (your notes) and **check the total energy** the whole system would have after the impulse had finished propagating. **I sent a single impulse with a strength of `5` into the cloud with the greatest number of outgoing connections** and followed its charge as it spread through the sky.

The starting cloud gained `5` energy and sent a separate impulse with a strength of `4` through each of its outgoing connections. Every cloud reached by an impulse gained energy equal to the impulse's current strength. If that strength was greater than 1, the cloud then sent a separate copy of the impulse through each of its outgoing connections, reducing its strength by 1. An impulse with a strength of `1` increased the cloud's energy, but travelled no further.

Different paths were treated separately. **If several impulses reached the same cloud, that cloud gained energy from every one of them.** Each impulse propagated independently. If a cloud received two impulses of the same strength, it sent two separate impulses through each outgoing connection. An impulse could also return to a cloud that it had already visited through a different connection. A connection could even lead directly back to the same cloud.

Example based on the following notes:

```
id=1, energy=5, connections=2,5,7
id=2, energy=9, connections=6,7
id=3, energy=3, connections=4
id=4, energy=8, connections=3
id=5, energy=1, connections=6,7
id=6, energy=7, connections=1,7
id=7, energy=6, connections=5
```

Cloud `1` has three outgoing connections, while every other cloud has no more than two, so cloud `1` is used as the starting cloud. First, `5` energy is added to cloud `1`. Then, impulses with a strength of `4` are sent to clouds `2`, `5` and `7`. These three impulses continue independently. For example, clouds `2` and `5` both send an impulse with a strength of `3` to cloud `6`. As a result, `3` energy is added to cloud `6` twice.

The total energy added during each stage is:

```
Strength 5:
cloud 1 received 1 impulse: +5

Strength 4:
cloud 2 received 1 impulse: +4
cloud 5 received 1 impulse: +4
cloud 7 received 1 impulse: +4

Strength 3:
cloud 5 received 1 impulse: +3
cloud 6 received 2 impulses: +6
cloud 7 received 2 impulses: +6

Strength 2:
cloud 1 received 2 impulses: +4
cloud 5 received 2 impulses: +4
cloud 6 received 1 impulse: +2
cloud 7 received 3 impulses: +6

Strength 1:
cloud 1 received 1 impulse: +1
cloud 2 received 2 impulses: +2
cloud 5 received 5 impulses: +5
cloud 6 received 2 impulses: +2
cloud 7 received 5 impulses: +5
```

Clouds `3` and `4` are never reached by any impulse, so their energy does not change.

After the propagation finishes, the clouds have the following energy:

```
Cloud 1: 5 + 10 = 15
Cloud 2: 9 + 6 = 15
Cloud 3: 3 + 0 = 3
Cloud 4: 8 + 0 = 8
Cloud 5: 1 + 16 = 17
Cloud 6: 7 + 10 = 17
Cloud 7: 6 + 21 = 27
```

The total energy of all the clouds is therefore:
15 + 15 + 3 + 8 + 17 + 17 + 27 = **102**.

What was the total energy of all the clouds after the impulse had finished propagating?

Your notes for this part:
Copy Open Download

Part I solved with answer: 253
Check your progress

Part II

I decided to **practise once more using a slightly larger arrangement of clouds** (your notes). This time, I used the full impulse strength of `10`. The impulse still began in the cloud with the greatest number of outgoing connections. That cloud gained `10` energy, and the impulse then propagated through successive stages with strengths of `9`, `8`, `7` and so on, until the final stage used `1`.

Example based on the following notes:

```
id=1, energy=5, connections=2,5,7
id=2, energy=9, connections=6,7
id=3, energy=3, connections=4
id=4, energy=8, connections=3
id=5, energy=1, connections=6,7
id=6, energy=7, connections=1,7
id=7, energy=6, connections=5
```

As in Part I, cloud `1` has the greatest number of outgoing connections, so it is used as the starting cloud and receives the initial impulse. The total energy added during each stage is listed **here**. After all ten stages, the clouds have the following energy:

```
Cloud 1: 5 + 147 = 152
Cloud 2: 9 + 83 = 92
Cloud 3: 3 + 0 = 3
Cloud 4: 8 + 0 = 8
Cloud 5: 1 + 343 = 344
Cloud 6: 7 + 242 = 249
Cloud 7: 6 + 462 = 468
```

The total energy of all the clouds is therefore:
152 + 92 + 3 + 8 + 344 + 249 + 468 = **1316**

```
id=1, energy=5, connections=2,3,5,7
id=2, energy=9, connections=1,3,4,5,6,7
id=3, energy=3, connections=1,2,4,6,7
id=4, energy=8, connections=1,2,7
id=5, energy=1, connections=2,3,4,6
id=6, energy=7, connections=1,2,3,4
id=7, energy=6, connections=1,2,4,6
```

In this example, cloud `2` has the greatest number of outgoing connections, so it is used as the starting cloud and receives the initial impulse. The total energy added during each stage is listed **here**.

The total energy of all the clouds after the impulse finishes propagating is: **1214633**

What was the total energy of all the clouds after the impulse had finished propagating?

Your notes for this part:
Copy Open Download

Part II solved with answer: 125719396
Check your progress

Part III

Sarumonk had no intention of waiting for me to finish my calculations. Before I could even raise my staff, he fired his energy impulse into the clouds. I watched the charge spread across the sky until one of the clouds flashed brightly and burst apart. That was when I remembered one rather important rule of lightning duels.

If a cloud reached an energy of at least `1000000`, it exploded. An exploded cloud disappeared from the system, so its energy was no longer included in the total energy of the remaining clouds.

An explosion also released another burst of energy. **After exploding, the cloud sent a new impulse with a strength of `10` through each of its outgoing connections.** Those new impulses propagated exactly as before, decreasing to 9, 8, 7 and so on.

Understanding the order of events was crucial.

- First, all clouds received and passed on the impulses of the current stage as usual.
- After that, every cloud that had reached the limit exploded.
- Each explosion generated an additional impulse with a strength of `10` for every outgoing connection. These new impulses joined the ones already passed on, as if they had all been emitted at exactly the same moment, and travelled together during the next stage.
- Once a cloud had exploded, it could no longer receive or pass any later impulses.

I still had only one shot, with a strength of `10`, but the cloud with the greatest number of outgoing connections was no longer an obvious choice. Every cloud that exploded disappeared, taking its energy out of the final total. With explosions able to trigger further impulses, I needed to find a new strategy.

Using the arrangement of clouds (your notes), I had to determine **which cloud to hit with my initial impulse so that the total energy of all clouds remaining after the entire propagation was as high as possible.**

Example based on the following notes:

```
id=1, energy=5, connections=2,5,7
id=2, energy=9, connections=6,7
id=3, energy=3, connections=4
id=4, energy=8, connections=3
id=5, energy=1, connections=6,7
id=6, energy=7, connections=1,7
id=7, energy=6, connections=5
```

To make the explosions easier to follow in this small example, the explosion limit is reduced to `100`.

When cloud `2` is used as the starting cloud, cloud `7` eventually reaches `135` energy and explodes. Its explosion sends a new impulse with a strength of `10` to cloud `5`.

During the following stage, clouds `5` and `6` also reach the explosion limit. All impulses in that stage are processed first. The two clouds then explode and generate additional impulses for the next stage.

After the entire propagation finishes, clouds `5`, `6` and `7` have exploded. The remaining clouds have the following energy:

```
Cloud 1: 94
Cloud 2: 82
Cloud 3: 3
Cloud 4: 8
```

Their total energy is therefore:
94 + 82 + 3 + 8 = **187**

The stage-by-stage process is listed **here**.

Every other starting cloud results in a lower total energy:

```
cloud 1: 108
cloud 2: 187 <- best
cloud 3: 94
cloud 4: 94
cloud 5: 101
cloud 6: 11
cloud 7: 159
```

```
For easier debugging of this rather complicated task,
one more example is available here.

The calculation file contains the results by starting cloud,
as well as the detailed process for the best choice.
```

What was the highest total energy of all the clouds remaining after sending the initial impulse into the best possible cloud?

Your notes for this part:
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Part III solved with answer: 30654897
Check your progress

Puzzle solved! Don't stop now!

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